State Space Methods for Flight Vehicles

AE 695

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Fields

- **Set**: A collection of elements or objects
- Examples: Set of all real or complex numbers

Fields

A field $\mathbb{F}$ is a set of elements, referred to as *scalars*, together with two operations of addition and multiplication for which the following axioms hold:

1. $\forall\ a, b \in \mathbb{F}$, there is always a unique sum $a + b \in \mathbb{F}$ and a unique product $a \cdot b \in \mathbb{F}$

$$\left.\begin{aligned} a + b &= b + a \\ ab &= ba \end{aligned}\right\} \text{ commutative laws}$$

2. for any three elements, $a, b, c \in \mathbb{F}$

$$\left.\begin{aligned} a + (b + c) &= (a + b) + c \\ a(bc) &= (ab)c \end{aligned}\right\} \text{ associative laws}$$

$$\left.a(b + c) = ab + ac\right\} \text{ distributive law}$$



Fields

3. $\mathbb{F}$ contains a zero element, 0, and a unity element, 1, such that

$$a + 0 = a,\ 1 \cdot a = a\ \forall a \in \mathbb{F}$$

4. $\forall a \in \mathbb{F},\ \exists\, b \in \mathbb{F}$ such that $a + b = 0$ (additive inverse)
5. $\forall a \in \mathbb{F} \setminus \{0\},\ \exists\, b \in \mathbb{F}$ such that $a \cdot b = 1$ (multiplicative inverse)

Example (Assuming usual rules of addition and multiplication)

- The set of all real numbers, $\mathbb{R}$ (**Yes!**)
- The set of all complex numbers, $\mathbb{C}$ (**Yes!**)
- The set of all positive real numbers, $\mathbb{R}_+$ (**No!**)
- The set of all rational functions, $\mathbb{R}(s)$ with real coefficients (**Yes!**)
- $\mathcal{F}$ may consist of either a finite or infinite number of elements.
- Will the set $\{0, 1\}$ form a field?
- No, with the usual definitions of addition and multiplication



Field: Examples

- Will the answer differ if the addition and multiplication rules are changed?
- Addition and multiplication rules

$$\boxed{0+0=1+1=0,\ 1+0=1,\ 0\cdot 1=0\cdot 0=0,\ 1\cdot 1=1}$$

- **A field of binary numbers**
- Set of 2×2 matrices of the form

$$\begin{bmatrix} x & -y \\ y & x \end{bmatrix},\ x,\ y \in \mathbb{R}$$

- What are the element 0 and 1 for this field?

$$0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},\quad 1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Will the set of all 2×2 matrices form a field? No
- What about a set of positive real numbers, integers, and polynomials? No



Vectors

A vector $\mathbf{x}$ is an ordered set of $n-$tuples of elements with addition and multiplication by a scalar defined in the usual way.

- Let $\mathbf{x} = [x_1 \; x_2 \; \cdots \; x_n]^\top$ and $\mathbf{y} = [y_1 \; y_2 \; \cdots \; y_n]^\top$, then

$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

$$\alpha\mathbf{x} = \alpha \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{bmatrix}, \text{ where } \alpha \in \mathbb{F}$$



Vector Spaces

A *vector space* over a field $\mathbb{F}$ is a set $\mathbb{V}$, denoted as $\mathbb{V}(\mathbb{F})$. The elements of $\mathbb{F}$ are called scalars, whereas the elements of $\mathbb{V}$ are called vectors. Together with two operations of vector addition and scalar multiplication, the following axioms hold:

1. $\forall\ \mathbf{x}, \mathbf{y} \in \mathbb{V}$, there is a unique sum $\mathbf{x} + \mathbf{y} \in \mathbb{V}$ (**closed under addition**)

$$\left.\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}\right\} \text{ commutative law}$$

2. $\forall\ \mathbf{x} \in \mathbb{V}$ and $\alpha \in \mathbb{F}$, there is a unique product $\alpha\mathbf{x} \in \mathbb{V}$ (**closed under scalar multiplication**)
3. $\forall\ \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{V}$

$$\left.(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})\right\} \text{ associative law}$$

4. $\forall\ \mathbf{x}, \mathbf{y} \in \mathbb{V}$ and $\alpha \in \mathbb{F}$,

$$\left.\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}\right\} \text{ distributive law}$$



Vector Spaces

5. $\forall\ \alpha, \beta \in \mathbb{F}$ and $\forall\ \mathbf{x} \in \mathbb{V}$,

$$\left.\alpha(\beta\mathbf{x}) = \alpha\beta\mathbf{x}\right\} \text{ associative law}$$

$$\left.(\alpha+\beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}\right\} \text{ distributive law}$$

6. $\mathbb{V}$ contains a null vector $\mathbf{0}$ such that $\forall\ \mathbf{x} \in \mathbb{V},\ \mathbf{x} + \mathbf{0} = \mathbf{x}$

7. $\forall\ \mathbf{x} \in \mathbb{V},\ \exists - \mathbf{x} \in \mathbb{V}$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

8. $\forall\ \mathbf{x} \in \mathbb{V}$, $1 \cdot \mathbf{x} = \mathbf{x}$, where $1 \in \mathbb{F}$

Example

- Will a field $\mathcal{F}$ form vector space over $\mathcal{F}$ with addition and multiplication defined on corresponding field?
- Yes, $(\mathbb{R}, \mathbb{R})$, $(\mathbb{C}, \mathbb{C})$
- What about $(\mathbb{C}, \mathbb{R})$, $(\mathbb{R}, \mathbb{C})$, $(\mathbb{R}(s), \mathbb{R}(s))$, $(\mathbb{R}(s), \mathbb{R})$, and $(\mathbb{R}, \mathbb{R}(s))$?



Vector Space: Examples

- Consider $\mathcal{F}^n$: $n-$tuples of scalars

$$\boldsymbol{x}_i = \begin{bmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{ni} \end{bmatrix}, \quad \boldsymbol{x}_i + \boldsymbol{x}_i = \begin{bmatrix} x_{1i} + x_{1j} \\ x_{2i} + x_{2j} \\ \vdots \\ x_{ni} + x_{nj} \end{bmatrix}, \quad \alpha \boldsymbol{x}_i = \begin{bmatrix} \alpha x_{1i} \\ \alpha x_{2i} \\ \vdots \\ \alpha x_{ni} \end{bmatrix}$$

- Is $(\mathcal{F}^n, \mathcal{F})$ a vector space? Yes
- $(\mathbb{R}^n, \mathbb{R})$, $(\mathbb{C}^n, \mathbb{C})$, $(\mathbb{R}^n(s), \mathbb{R}(s))$: Real, complex, and rational vector space
- Set of all polynomials of degree less than n with real coefficients

$$\mathbb{R}_n[s] = \sum_{i=0}^{n-1} \alpha_i s^i$$

- Vector addition and scalar multiplication

$$\sum_{i=0}^{n-1} \alpha_i s^i + \sum_{i=0}^{n-1} \beta_i s^i = \sum_{i=0}^{n-1} (\alpha_i + \beta_i) s^i, \quad \alpha \sum_{i=0}^{n-1} \alpha_i s^i = \sum_{i=0}^{n-1} (\alpha \alpha_i) s^i$$

- Is $(\mathbb{R}_n[s], \mathbb{R})$ a vector space? Yes; Is $(\mathbb{R}_n[s], \mathbb{R}(s))$ a vector space? No



Definition of Subspace

Subspace

Let $(\mathcal{X}, \mathcal{F})$ be a linear space and let $\mathcal{Y}$ be a subset of $\mathcal{X}$. Then $(\mathcal{Y}, \mathcal{F})$ is said to be a subspace of $(\mathcal{X}, \mathcal{F})$ if under the operations of $(\mathcal{X}, \mathcal{F})$, $\mathcal{Y}$ itself forms a vector space over $\mathcal{F}$.

- Many properties satisfied owing to the definitions of addition and multiplication over a vector space.
- **Need to check only remaining conditions**
- Closed under addition and scalar multiplication

$$\alpha_1 \boldsymbol{y}_1 + \alpha_2 \boldsymbol{y}_2 \in \mathcal{Y} \quad \forall\ \alpha_1, \alpha_2 \in \mathcal{F}, \boldsymbol{y}_1, \boldsymbol{y}_2 \in \mathcal{Y}$$

- What can you say about the line passing through origin in 2D real vector space? Is it a subspace of $(\mathbb{R}^2, \mathbb{R})$?

$$\begin{bmatrix} x_1 \\ \alpha x_1 \end{bmatrix}, \ \alpha \in \mathbb{R}$$

- Is $(\mathbb{R}^n, \mathbb{R})$ a subspace of $(\mathbb{C}^n, \mathbb{R})$, Yes



Linear Dependence

Linear dependence of a set of vectors

A set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2 \cdots, \boldsymbol{x}_n$ in linear space $(\mathcal{X}, \mathcal{F})$, is said to be linearly dependent iff there exist scalars $\alpha_1, \alpha_2, \cdots, \alpha_n \in \mathcal{F}$, not all zero, such that

$$\alpha_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2 + \cdots + \alpha_n \boldsymbol{x}_n = \boldsymbol{0}.$$

If the only set of α_i for which above holds is $\alpha_1 = 0, \alpha_2 = 0, \cdots, \alpha_n = 0$, then the set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$ is said to be linearly independent.

- For a dependent set of vectors, infinitely many choices of coefficients.
- Finding one such choice of coefficients suffices.
- What can we say about the l.i. of a set of vectors, $\boldsymbol{x}_1, \boldsymbol{x}_2, \cdots, \boldsymbol{x}_n$ if $\boldsymbol{x}_1 = \boldsymbol{0}$?
- Linearly dependent set of vectors
- What about linear dependence of a set of vectors, with only one vector $\boldsymbol{x}_1$?
- Linearly independent iff $\boldsymbol{x}_1 \neq \boldsymbol{0}$



Linear Dependence

- Define the notation

$$\alpha_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2 + \cdots + \alpha_n \boldsymbol{x}_n \triangleq \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n \end{bmatrix} \boldsymbol{\alpha}, \boldsymbol{\alpha} = \begin{bmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_n \end{bmatrix}^\top$$

Linear dependence of a set of vectors

A set of vectors $\boldsymbol{x}_1, \boldsymbol{x}_2 \cdots, \boldsymbol{x}_n$ in linear space $(\mathcal{X}, \mathcal{F})$, is said to be linearly independent iff the equation

$$\begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n \end{bmatrix} \boldsymbol{\alpha} = \boldsymbol{0} \implies \boldsymbol{\alpha} = \boldsymbol{0} \in \mathcal{F}^n,$$

where every component of $\boldsymbol{\alpha}$ is an element of $\mathcal{F}$.

- What about the linear dependence of the set of vectors given below?

$$\boldsymbol{x}_1 \triangleq \begin{bmatrix} \dfrac{1}{s+1} \\ \dfrac{1}{s+2} \end{bmatrix}, \quad \boldsymbol{x}_2 \triangleq \begin{bmatrix} \dfrac{s+2}{(s+1)(s+3)} \\ \dfrac{1}{s+3} \end{bmatrix}$$

- Linearly dependent in $\left(\mathbb{R}^2(s), \mathbb{R}(s)\right)$, but independent in $\left(\mathbb{R}^2(s), \mathbb{R}\right)$



Linear Dependence: Dimension

- **Linear dependence**: At least one vector as a linear combination of other vectors.
- May not be possible to express all vectors in a similar manner.

Dimension of a vector space

The maximal number of l.i vectors in a linear space $(\mathcal{X}, \mathcal{F})$.

- What is the dimension of the vector space $(\mathbb{R}^2, \mathbb{R})$? 2
- What would be the dimension of the vector space of all real-valued piece-wise continuous functions over $(-\infty, \infty)$? ∞ How?

Definition of the basis of vector space

A set of linearly independent vectors of a linear space $(\mathcal{X}, \mathcal{F})$ if every vector in $\mathcal{X}$ can be expressed as a unique linear combination of these vectors.



Linear Dependence: Basis

In an n-dimensional vector space, any set of n linearly independent vectors qualifies as a basis.

- Assume $\boldsymbol{e}_i \forall\ i = 1, 2, \ldots, n$ be n l.i vectors in $\mathcal{X}$.
- With arbitrary vector $\boldsymbol{x}$, the set $\{\boldsymbol{x}, \boldsymbol{e}_i \,\forall\, i = 1, 2, \ldots, n\}$ are l.d.
- Thus, $\exists\, \alpha_i \,\forall\, i = 0, 1, \cdots, n$ not all zero s.t.

$$\alpha_0 \boldsymbol{x} + \sum \alpha_i \boldsymbol{e}_i = \boldsymbol{0}$$

- For $\alpha_0 = 0$, due to the linear independence of vectors,

$$\sum \alpha_i \boldsymbol{e}_i = \boldsymbol{0} \implies \alpha_i = 0, \quad \text{Contradictions !!}$$

- With $\beta_i \triangleq -\alpha_i/\alpha_0$, we have

$$\boxed{\boldsymbol{x} = \sum_{i=1}^{n} \beta_i \boldsymbol{e}_i}$$



Linear Dependence: Basis

- We need to show that this linear combination is unique.
- Assume that we have another linear combination for the same vector

$$\boldsymbol{x} = \sum_{i=1}^{n} \tilde{\beta}_i \boldsymbol{e}_i = \sum_{i=1}^{n} \beta_i \boldsymbol{e}_i \implies \boxed{\boldsymbol{0} = \sum_{i=1}^{n} \left(\beta_i - \tilde{\beta}_i\right) \boldsymbol{e}_i}$$

- Owing to the linear independent nature of $\boldsymbol{e}_i$, one can write

$$\beta_i = \tilde{\beta}_i \ \forall \ i = 1, 2, \ldots, n.$$

Representation of vector

In $n-$dimensional vector space, with basis $\{\boldsymbol{e}_i \, \forall \, i = 1, 2, \ldots, n\}$, any vector $\boldsymbol{x}$ can be uniquely written w.r.t. chosen basis as

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \cdots & \boldsymbol{e}_n \end{bmatrix} \boldsymbol{\beta}, \quad \boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_1 & \boldsymbol{\beta}_1 & \cdots & \boldsymbol{\beta}_n \end{bmatrix}^\top$$

Representation of vector x in the chosen basis: β.



Linear Dependence: Basis

- **Example**: Two-dimensional real vector space
- **Freedom in choosing basis**: Direction and magnitude of vectors
- **Different representations for the same vector w.r.t. different bases**
- Consider a vector space $(\mathbb{R}(s), \mathbb{R})$.
- With the basis $\boldsymbol{e}_1 = s^3, \boldsymbol{e}_2 = s^2, \boldsymbol{e}_3 = s, \boldsymbol{e}_4 = 1$,

$$\boldsymbol{x} = 3s^3 + 2s^2 - 2s + 10 = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 & \boldsymbol{e}_4 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ -2 \\ 10 \end{bmatrix}$$

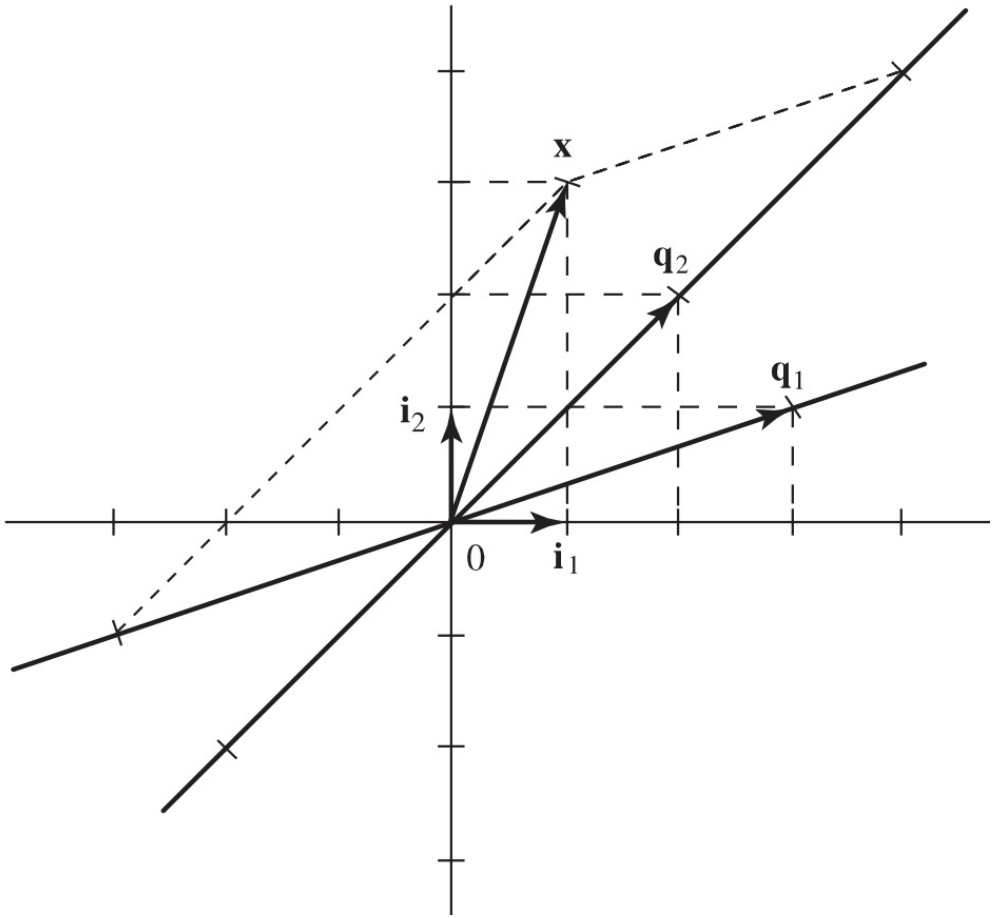
- With another basis $\bar{\boldsymbol{e}}_1 = s^3 - s^2, \bar{\boldsymbol{e}}_2 = s^2 - s, \bar{\boldsymbol{e}}_3 = s - 1, \bar{\boldsymbol{e}}_4 = 1$,

$$\boldsymbol{x} = 3s^3 + 2s^2 - 2s + 10 = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \bar{\boldsymbol{e}}_3 & \bar{\boldsymbol{e}}_4 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \\ 3 \\ 13 \end{bmatrix}$$



Linear Dependence: Basis

Obtain the representation of vector $\boldsymbol{x} = [1 \;\; 3]^\top$ w.r.t. bases $\boldsymbol{q}_1 = [3 \;\; 1]^\top$ and $\boldsymbol{q}_2 = [2 \;\; 2]^\top$, and also about $\boldsymbol{q}_2$ and $\boldsymbol{i}_2$.



- Draw from $\boldsymbol{x}$ two lines in parallel with $\boldsymbol{q}_2$ and $\boldsymbol{q}_1$, they intersect at $-\boldsymbol{q}_1$ and $2\boldsymbol{q}_2$.
- Representation of $\boldsymbol{x}$ w.r.t. $\boldsymbol{q}_1, \boldsymbol{q}_2$

$$\boldsymbol{x} = -\boldsymbol{q}_1 + 2\boldsymbol{q}_2$$

$$\boxed{\boldsymbol{x} = 0.5\boldsymbol{q}_2 + 2\boldsymbol{i}_2}$$

- Representations change w.r.t. different basis vectors set.

Linear Dependence: Basis

- How to interpret an array of n-tuple?
  - A vector independent of basis
  - A representation of vector w.r.t. some basis
- Define the basis vectors as

$$\boldsymbol{n}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \ \boldsymbol{n}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \cdots, \ \boldsymbol{n}_{n-1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}, \ \boldsymbol{n}_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

- Interpretation of any vector in this basis

$$\begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{n}_1 & \boldsymbol{n}_2 & \cdots & \boldsymbol{n}_n \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{bmatrix}$$

- Is there any relation between different representations of the same vector in different bases?



Linear Dependence: Change of Basis

- Representation of a vector $\boldsymbol{x}$ in $(\mathcal{X}, \mathcal{F})$ in two different bases

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \cdots & \boldsymbol{e}_n \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 \cdots & \bar{\boldsymbol{e}}_n \end{bmatrix} \bar{\boldsymbol{\beta}}$$

- Need of information on the representation of one basis w.r.t. other basis
- Assume the representation of $\boldsymbol{e}_i$ w.r.t. basis $\{\bar{\boldsymbol{e}}_i\}$

$$\boldsymbol{e}_i = \underbrace{\begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \cdots & \bar{\boldsymbol{e}}_n \end{bmatrix}}_{\boldsymbol{E}} \underbrace{\begin{bmatrix} p_{1i} \\ p_{2i} \\ \cdots \\ p_{ni} \end{bmatrix}}_{\boldsymbol{p}_i} \triangleq \boldsymbol{E}\boldsymbol{p}_i$$

- Using matrix notation,

$$\begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \ldots & \boldsymbol{e}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{E}\boldsymbol{p}_1 & \boldsymbol{E}\boldsymbol{p}_2 & \ldots & \boldsymbol{E}\boldsymbol{p}_n \end{bmatrix} = \boldsymbol{E} \underbrace{\begin{bmatrix} \boldsymbol{p}_1 & \boldsymbol{p}_2 & \ldots & \boldsymbol{p}_n \end{bmatrix}}_{\boldsymbol{P}}$$

$$\begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \ldots & \boldsymbol{e}_n \end{bmatrix} = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \cdots & \bar{\boldsymbol{e}}_n \end{bmatrix} \boldsymbol{P}$$



Linear Dependence: Change of Basis

- Vector $\boldsymbol{x}$ can be rewritten as

$$\boldsymbol{x} = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 \cdots & \boldsymbol{e}_n \end{bmatrix} \boldsymbol{\beta} = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \cdots & \bar{\boldsymbol{e}}_n \end{bmatrix} \boldsymbol{P}\boldsymbol{\beta} = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 \cdots & \bar{\boldsymbol{e}}_n \end{bmatrix} \bar{\boldsymbol{\beta}}$$

- Owing to the uniqueness of representation of $\boldsymbol{x}$ w.r.t. basis $\{\bar{\boldsymbol{e}}_i\}$

$$\boxed{\bar{\boldsymbol{\beta}} = \boldsymbol{P}\boldsymbol{\beta}}$$

- **The ith columns of $\boldsymbol{P}$**: Representation of $\boldsymbol{e}_i$ w.r.t. the basis $\{\bar{\boldsymbol{e}}_i\}$
- Similarly, using the representation of $\bar{\boldsymbol{e}}_i$ w.r.t. $\{\boldsymbol{e}_i\}$

$$\boxed{\boldsymbol{\beta} = \boldsymbol{Q}\bar{\boldsymbol{\beta}}}$$

- **The ith columns of $\boldsymbol{Q}$**: Representation of $\bar{\boldsymbol{e}}_i$ w.r.t. $\{\boldsymbol{e}_i\}$
- Using both relations

$$\bar{\boldsymbol{\beta}} = \boldsymbol{P}\boldsymbol{Q}\bar{\boldsymbol{\beta}} \;\forall\; \bar{\boldsymbol{\beta}} \implies \boxed{\boldsymbol{P}\boldsymbol{Q} = \boldsymbol{I}, \quad \boldsymbol{P} = \boldsymbol{Q}^{-1}}$$



Linear Dependence: Change of Basis

- Consider the bases given by

$$\boldsymbol{e}_1 = s^3, \boldsymbol{e}_2 = s^2, \boldsymbol{e}_3 = s, \boldsymbol{e}_4 = 1$$

$$\bar{\boldsymbol{e}}_1 = s^3 - s^2, \bar{\boldsymbol{e}}_2 = s^2 - s, \bar{\boldsymbol{e}}_3 = s - 1, \bar{\boldsymbol{e}}_4 = 1$$

- Relation between bases

$$\begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 & \boldsymbol{e}_4 \end{bmatrix} = \begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \bar{\boldsymbol{e}}_3 & \bar{\boldsymbol{e}}_4 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}}_{\boldsymbol{P}}$$

$$\begin{bmatrix} \bar{\boldsymbol{e}}_1 & \bar{\boldsymbol{e}}_2 & \bar{\boldsymbol{e}}_3 & \bar{\boldsymbol{e}}_4 \end{bmatrix} = \begin{bmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 & \boldsymbol{e}_4 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -1 \end{bmatrix}}_{\boldsymbol{Q}}$$

- To verify as homework $\boxed{\boldsymbol{PQ} = \boldsymbol{I},\ \bar{\boldsymbol{\beta}} = \boldsymbol{P}\boldsymbol{\beta}}$



Orthonormal Basis

A vector $\boldsymbol{x}$ is said to be normalized if its length is unity.

Two vectors $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ are said to be orthogonal if $\boxed{\boldsymbol{x}_1^\top \boldsymbol{x}_2 = \boldsymbol{x}_2^\top \boldsymbol{x}_1 = 0}$.

A set of vectors $\boldsymbol{x}_i \forall\ i = 1, 2, \ldots, m$, is said to be orthonormal if

$$\boldsymbol{x}_i^\top \boldsymbol{x}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

- For a square matrix $\boldsymbol{A}$ with all orthonormal columns $\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I}_n$
- For a nonsquare $(n \times m)$ matrix $\boldsymbol{A}$ with all orthonormal columns

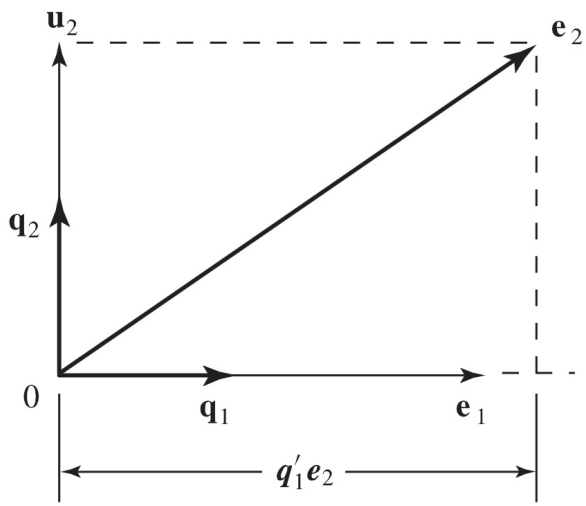
$$\boxed{\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{I}_m, \quad \boldsymbol{A}\boldsymbol{A}^\top \neq \boldsymbol{I}_n \text{ (In general)}}$$

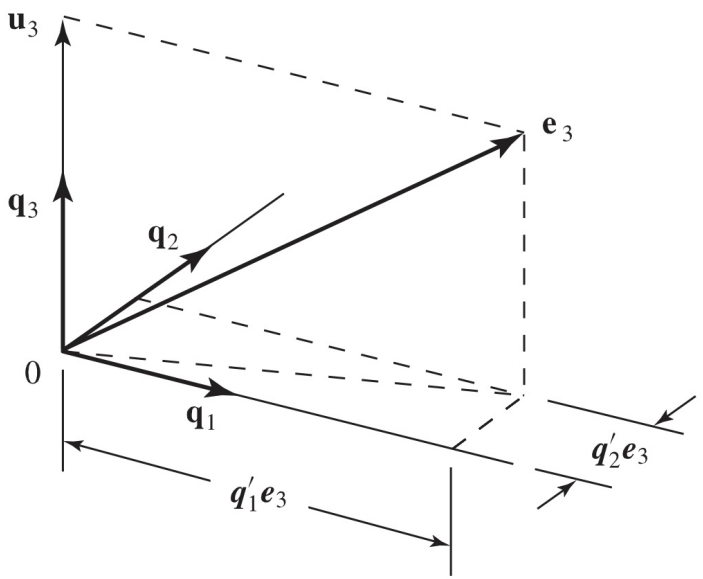
- How to obtain an orthonormal basis from a given set of linearly independent vectors, $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_m\}$?



Orthonormal Basis: Schmidt Orthonormalization

$$
\begin{aligned}
\boldsymbol{u}_1 &= \boldsymbol{e}_1, & \boldsymbol{q}_1 &:= \frac{\boldsymbol{u}_1}{\|\boldsymbol{u}_1\|} \\
\boldsymbol{u}_2 &= \boldsymbol{e}_2 - \left(\boldsymbol{q}_1^\top \boldsymbol{e}_2\right) \boldsymbol{q}_1, & \boldsymbol{q}_2 &:= \frac{\boldsymbol{u}_2}{\|\boldsymbol{u}_2\|} \\
&\vdots & & \\
\boldsymbol{u}_m &= \boldsymbol{e}_m - \sum_{k=1}^{m-1} \left(\boldsymbol{q}_k^\top \boldsymbol{e}_m\right) \boldsymbol{q}_k, & \boldsymbol{q}_m &:= \frac{\boldsymbol{u}_m}{\|\boldsymbol{u}_m\|}
\end{aligned}
$$



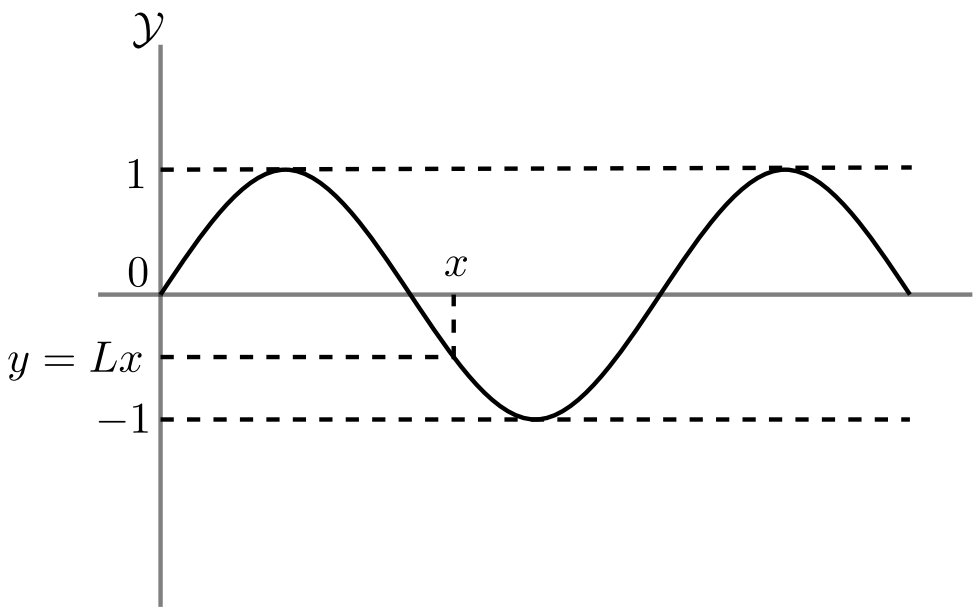


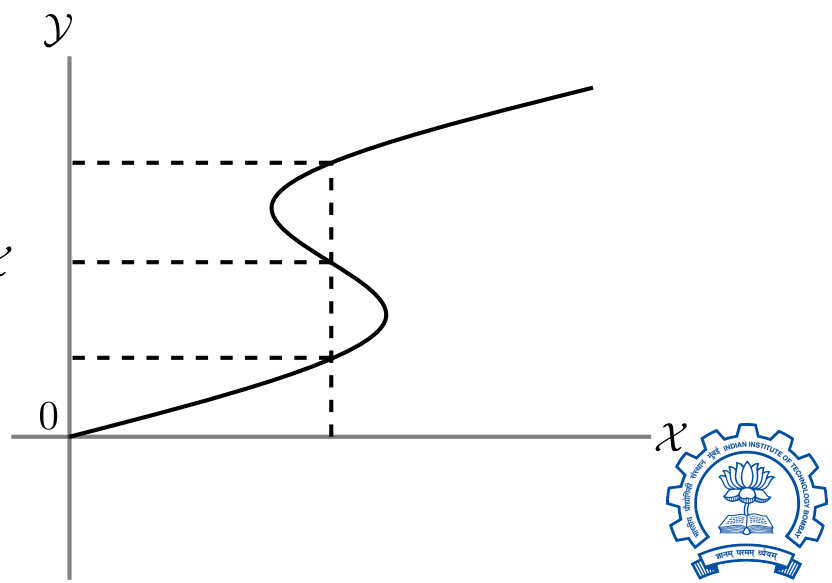
Linear Operators

- How to define a function?
- Defined as a rule of assignment by which we assign each element of $\mathcal{X}$ one and only one element in $\mathcal{Y}$ for two given sets $\mathcal{X}, \mathcal{Y}$.
- **Notation**: $f : \mathcal{X} \to \mathcal{Y}$

$$y = f(x)$$

- **Domain**: The set $\mathcal{X}$ on which a function is defined.
- **Range**: The subset of $\mathcal{Y}$ that is assigned to some element of $\mathcal{X}$.





Linear Operators

- **Linear operator (function, mapping, transformation)**
- Associated set must be a linear space over the same field

Linear operator

A function $L : (\mathcal{X}, \mathcal{F}) \to (\mathcal{Y}, \mathcal{F})$ is said to be linear operator iff

$$L(\alpha_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2) = \alpha_1 L\boldsymbol{x}_1 + \alpha_2 L\boldsymbol{x}_2$$

for any vectors $\boldsymbol{x}_1, \boldsymbol{x}_2 \in \mathcal{X}$ and any scalars $\alpha_1, \alpha_2 \in \mathcal{F}$.

- What would be the reason for the requirement of the same field for $\mathcal{Y}$?
- To define $\alpha_1 L\boldsymbol{x}_1,\ \alpha_2 L\boldsymbol{x}_2$
- What can you say about the transformation that rotates a point in the geometric plane CCW by 90° w.r.t. the origin?
- Linear operator



Linear Operators

- Linear operators can be defined on both finite and infinite-dimensional vector spaces.
- Linear operator defined on finite-dimensional vector spaces has matrix representations.

Let $(\mathcal{X}, \mathcal{F})$ and $(\mathcal{Y}, \mathcal{F})$ be the n and m-dimensional vector spaces, respectively, over same field. Let $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n$ be the set of l.i. vectors in $\mathcal{X}$. Then, the linear operator $L : (\mathcal{X}, \mathcal{F}) \to (\mathcal{Y}, \mathcal{F})$ is uniquely determined by the n pairs of mappings $\boldsymbol{y}_i = L\boldsymbol{x}_i \forall\ i = 1, 2, \ldots, n$.

Furthermore, with respect to the basis $\{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ of $\mathcal{X}$, and a basis $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_m\}$ of $\mathcal{Y}$, L can be represented by an $m \times n$ matrix $\boldsymbol{A}$ with the coefficients in $\mathcal{F}$. The ith column of $\boldsymbol{A}$ is the representation of $\boldsymbol{y}_i$ w.r.t. the basis $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_m\}$.



Linear Operators

- For arbitrary vector $\boldsymbol{x} \in \mathcal{X}$, one can write

$$\boldsymbol{x} = \alpha_1 \boldsymbol{x}_1 + \alpha_2 \boldsymbol{x}_2 + \cdots + \alpha_n \boldsymbol{x}_n$$

- Using the linearity of operator L

$$L\boldsymbol{x} = \alpha_1 L\boldsymbol{x}_1 + \alpha_2 L\boldsymbol{x}_2 + \cdots + \alpha_n L\boldsymbol{x}_n = \alpha_1 \boldsymbol{y}_1 + \alpha_2 \boldsymbol{y}_2 + \cdots + \alpha_n \boldsymbol{y}_n$$

- Vector $L\boldsymbol{x}$ uniquely determined by $\boldsymbol{y}_i = L\boldsymbol{x}_i \forall\ i = 1, 2, \ldots, n.$

- Representation of $\boldsymbol{y}_i$ w.r.t. basis $\{\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_m\}$

$$\boldsymbol{y}_i = \begin{bmatrix} \boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m \end{bmatrix} \begin{bmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{mi} \end{bmatrix} \quad \forall\ i = 1, 2, \ldots, n$$

$$L \begin{bmatrix} \boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n \end{bmatrix} = \begin{bmatrix} \boldsymbol{y}_1 & \boldsymbol{y}_2 & \cdots & \boldsymbol{y}_n \end{bmatrix}$$



Linear Operators

- We can write

$$L\begin{bmatrix}\boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n\end{bmatrix} = \begin{bmatrix}\boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m\end{bmatrix} \underbrace{\begin{bmatrix} a_{11} & a_{12}\cdots & a_{1n} \\ a_{21} & a_{22}\cdots & a_{2n} \\ \vdots & & \\ a_{m1} & a_{m2}\cdots & a_{mn}\end{bmatrix}}_{\boldsymbol{A}}$$

- The ith column of $\boldsymbol{A}$: Representation of $\boldsymbol{y}_i$ w.r.t. basis of $\mathcal{Y}$
- Linear operator $\boldsymbol{y} = L\boldsymbol{x}$

$$\begin{bmatrix}\boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m\end{bmatrix}\boldsymbol{\beta} = L\begin{bmatrix}\boldsymbol{x}_1 & \boldsymbol{x}_2 & \cdots & \boldsymbol{x}_n\end{bmatrix}\boldsymbol{\alpha}$$

$$\begin{bmatrix}\boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m\end{bmatrix}\boldsymbol{\beta} = \begin{bmatrix}\boldsymbol{u}_1 & \boldsymbol{u}_2 & \cdots & \boldsymbol{u}_m\end{bmatrix}\boldsymbol{A}\boldsymbol{\alpha}$$

- Owing to the uniqueness of representation: $\boxed{\boldsymbol{\beta} = \boldsymbol{A}\boldsymbol{\alpha}}$
- For the chosen bases, the linear operator represented as a matrix $\boldsymbol{A}$.



Linear Operators

- Matrix $\boldsymbol{A}$ relates $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, not vectors $\boldsymbol{x}$ and $\boldsymbol{y}$.
- **Linear Operator**: Different matrix representations for different bases
- What would be the relation if two different bases were used to represent such linear operators?
- Assume $\boldsymbol{x} \in \mathcal{X}$ has representations $\boldsymbol{\alpha}, \bar{\boldsymbol{\alpha}}$ w.r.t. the bases $\{\boldsymbol{e}_i\}$ and $\{\bar{\boldsymbol{e}}_i\}$.
- Representation of $\boldsymbol{y} = L\boldsymbol{x}$ in these bases: $\boldsymbol{\beta}$ and $\bar{\boldsymbol{\beta}}$

$$\bar{\boldsymbol{\alpha}} = \boldsymbol{P}\boldsymbol{\alpha}, \quad \bar{\boldsymbol{\beta}} = \boldsymbol{P}\boldsymbol{\beta}$$

$$\boxed{\bar{\boldsymbol{\beta}} = \bar{\boldsymbol{A}}\bar{\boldsymbol{\alpha}}}, \qquad \bar{\boldsymbol{\beta}} = \boldsymbol{P}\boldsymbol{\beta} = \boldsymbol{P}\boldsymbol{A}\boldsymbol{\alpha} = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}\bar{\boldsymbol{\alpha}}$$

- Owing to the uniqueness of representation w.r.t. specific basis

$$\bar{\boldsymbol{A}}\bar{\boldsymbol{\alpha}} = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1}\bar{\alpha} \implies \boxed{\bar{\boldsymbol{A}} = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}}$$

$$\boxed{\boldsymbol{A} = \boldsymbol{P}^{-1}\bar{\boldsymbol{A}}\boldsymbol{P} = \boldsymbol{Q}\bar{\boldsymbol{A}}\boldsymbol{Q}^{-1}}$$

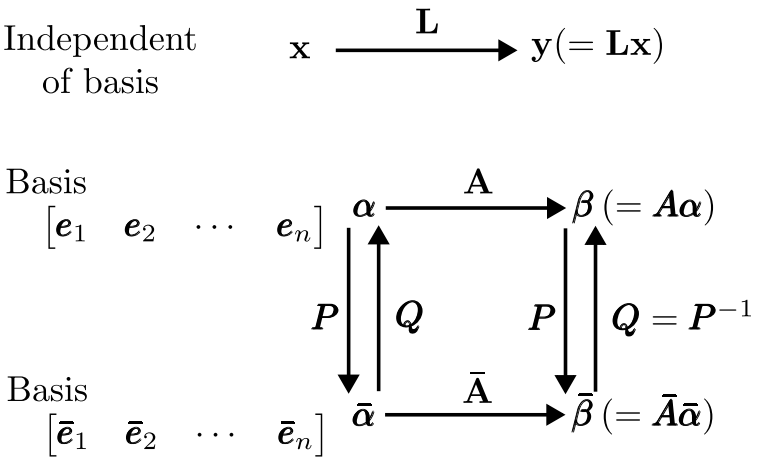


Similarity Transformation

Two matrices $\boldsymbol{A}$ and $\bar{\boldsymbol{A}}$ are said to be similar if there exists a nonsingular transformation matrix $\boldsymbol{P}$ satisfying

$$\bar{\boldsymbol{A}} = \boldsymbol{P}\boldsymbol{A}\boldsymbol{P}^{-1} = \boldsymbol{Q}^{-1}\boldsymbol{A}\boldsymbol{Q}, \quad \boldsymbol{A} = \boldsymbol{P}^{-1}\bar{\boldsymbol{A}}\boldsymbol{P} = \boldsymbol{Q}\bar{\boldsymbol{A}}\boldsymbol{Q}^{-1}, \quad \boldsymbol{Q} = \boldsymbol{P}^{-1}$$

- All the matrix representations of the same operator w.r.t. different bases are similar.



$$\mathbf{A} = \begin{bmatrix} i^{\text{th}} \text{ column:} \\ \text{Representation of} \\ L\boldsymbol{e}_i \text{ w.r.t. the basis} \\ \{\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n\} \end{bmatrix}, \quad \boldsymbol{P} = \begin{bmatrix} i^{\text{th}} \text{ column:} \\ \text{Representation of} \\ \boldsymbol{e}_i \text{ w.r.t. the basis} \\ \{\bar{\boldsymbol{e}}_1, \bar{\boldsymbol{e}}_2, \cdots, \bar{\boldsymbol{e}}_n\} \end{bmatrix}$$

$$\bar{\mathbf{A}} = \begin{bmatrix} i^{\text{th}} \text{ column:} \\ \text{Representation of} \\ L\bar{\boldsymbol{e}}_i \text{ w.r.t. the basis} \\ \{\bar{\boldsymbol{e}}_1, \bar{\boldsymbol{e}}_2, \cdots, \bar{\boldsymbol{e}}_n\} \end{bmatrix}, \quad \boldsymbol{Q} = \begin{bmatrix} i^{\text{th}} \text{ column:} \\ \text{Representation of} \\ \bar{\boldsymbol{e}}_i \text{ w.r.t. the basis} \\ \{\boldsymbol{e}_1, \boldsymbol{e}_2, \cdots, \boldsymbol{e}_n\} \end{bmatrix}$$

Reference

Reference

- **Chen, Chi-Tsong, "Linear System Theory and Design", Oxford University Press, 3rd Edition, 1999.**

